PROFESSIONAL DIPLOMA IN

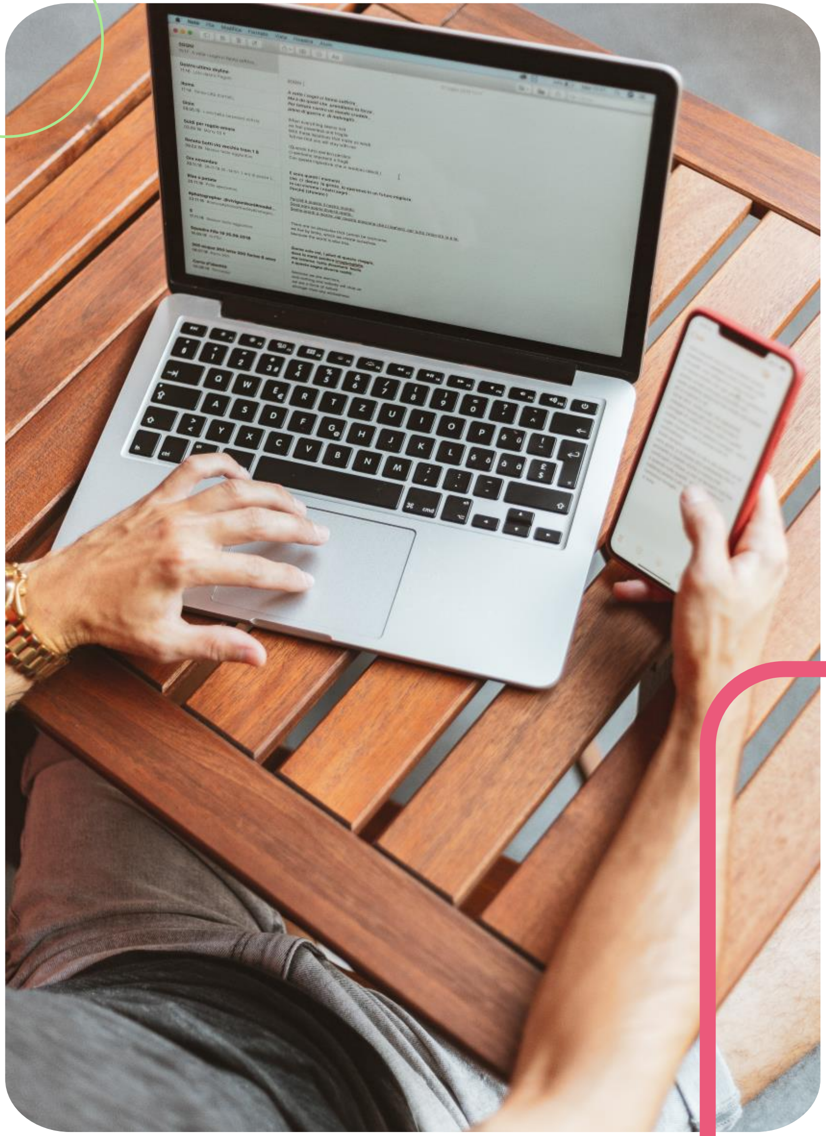
DATA ANALYSIS

Lesson 2: Exploring Data

$50 Billion

DID YOU KNOW?

Bad data costs the United States businesses an estimated $50 billion annually!



Data Types

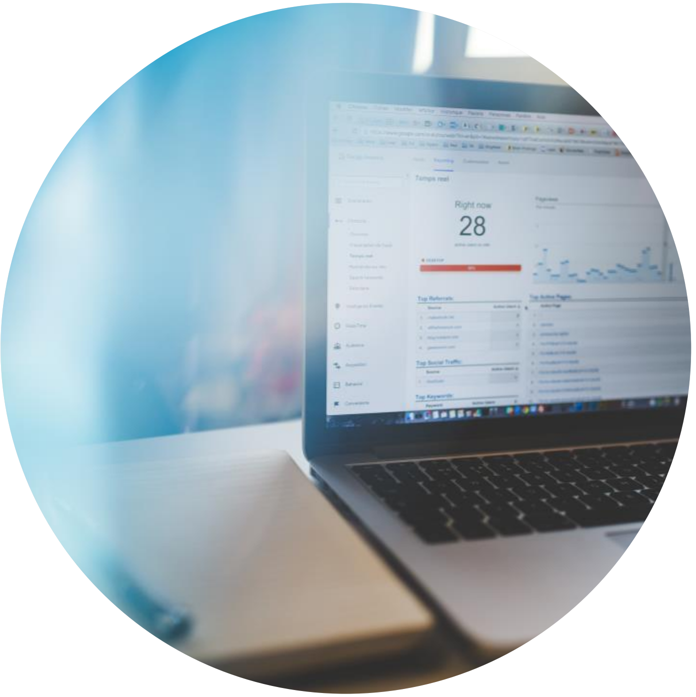
Descriptive Stats

Visualizations

Lesson Objectives

Lesson 2

Recap Lesson 1:
Intro to Data Analysis



Intro to
data analysis

Intro to
data

Importing and
cleaning data

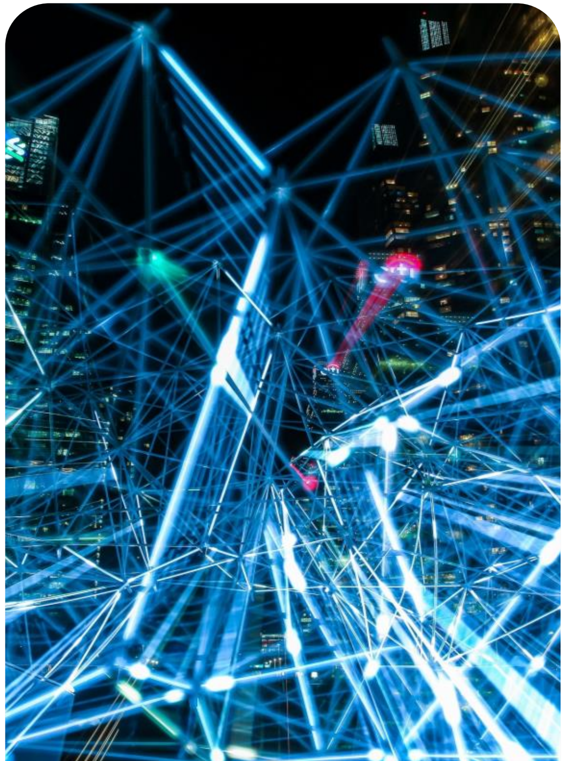
Data types

Data can be...

(quick recap)

Quantitative

Non-numerical, subjective, used to answer the why, used to generate hypothesis



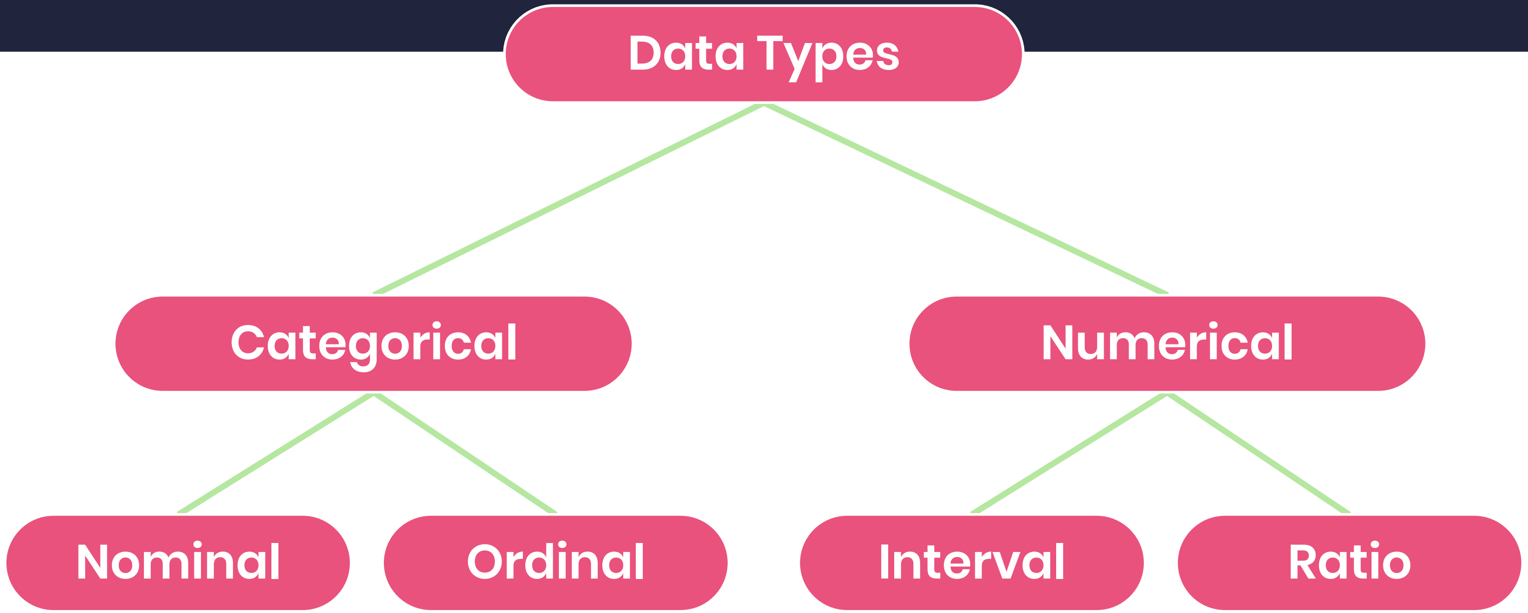
Qualitative

Numerical, objective, used to answer hypothesis, statistics to draw conclusions

Why do I need to know more?

- Different data type analysed differently
- Different data types uses different visualisations

Breaking data down



Categorical data

Data with a limited, fixed number of possible values

Nominal

Naming or labelling variables

Which Shaw Academy courses are you taking?

- (selected) Data Analytics
- Excel
- Technology
- Other

Ordinal

Categorical data with an order

How happy are you with this course this far?

- (selected) Very happy
- Happy
- OK
- Unhappy
- Very unhappy

Numerical data

Discrete

Finite possible values, Distinct and separate values, counted, only certain values

How many lessons are in this module?

- 2
- 4
- 8
- Other

Continuous

Infinite possible values in selected range

How long did it take you to complete the quiz?

- 1 minute
- 1,1 minutes
- 1,2 minutes
- 1,3 minutes
- 1,4 minutes
- 1,5 minutes
- Longer than 1,5 minutes

Continuous numerical data

Interval

What grade did you receive in your quiz?

- (●) A
- ( ) B
- ( ) C
- ( ) Other

Ratio

How long did it take you to complete the quiz?

- ( ) 1 minute
- ( ) 1,1 minutes
- ( ) 1,2 minutes
- ( ) 1,3 minutes
- ( ) 1,4 minutes
- ( ) 1,5 minutes
- (●) Longer than 1,5 minutes

Descriptive statistics

Why descriptive statistics?

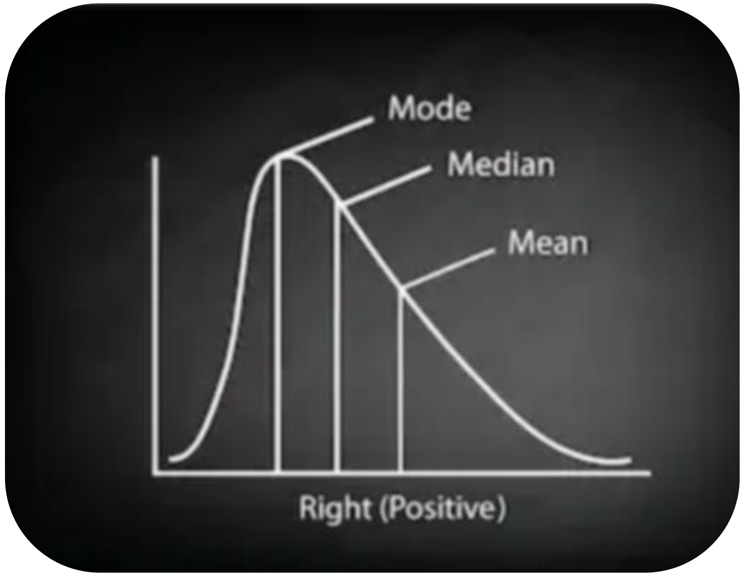
- Simpler interpretation of data
- More meaningful
- Highlight relationships

Measures of...

(a look into 2 of the ways of describing data)

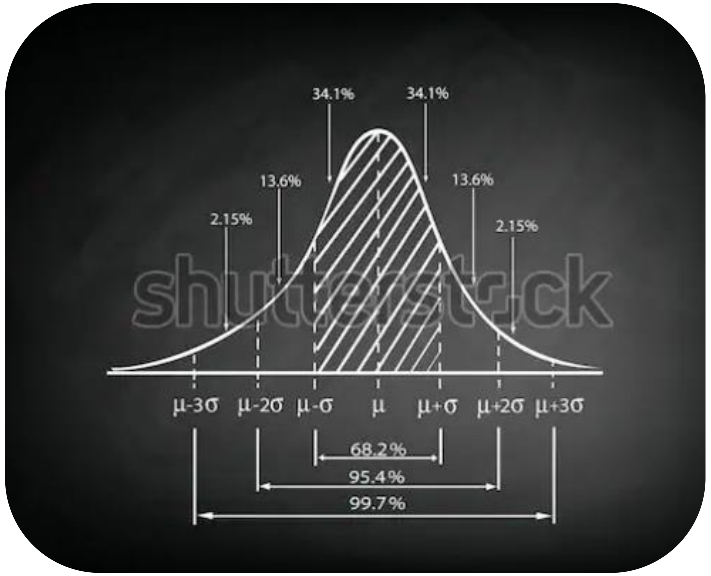
Central tendency

Finite possible values, distinct and separate values, counted, only certain values



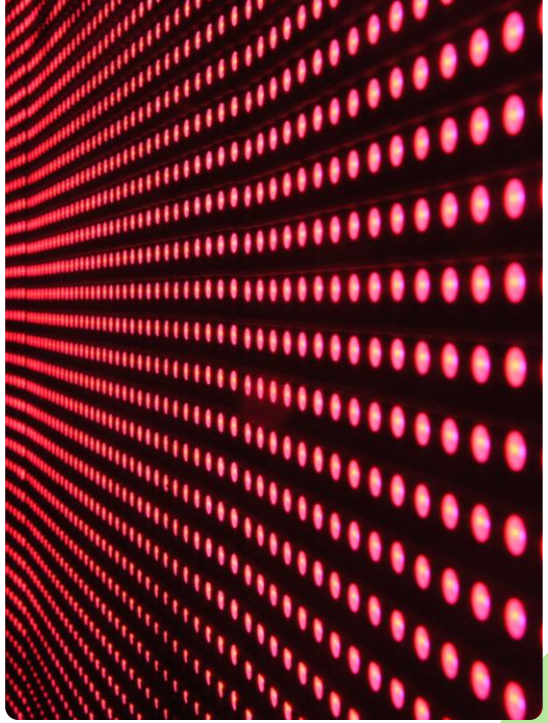
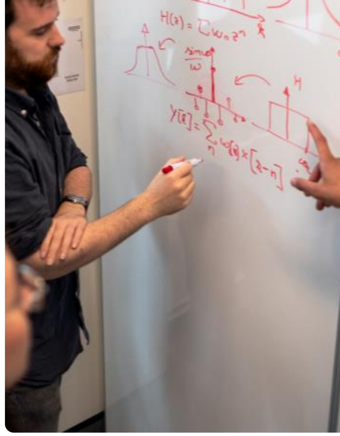
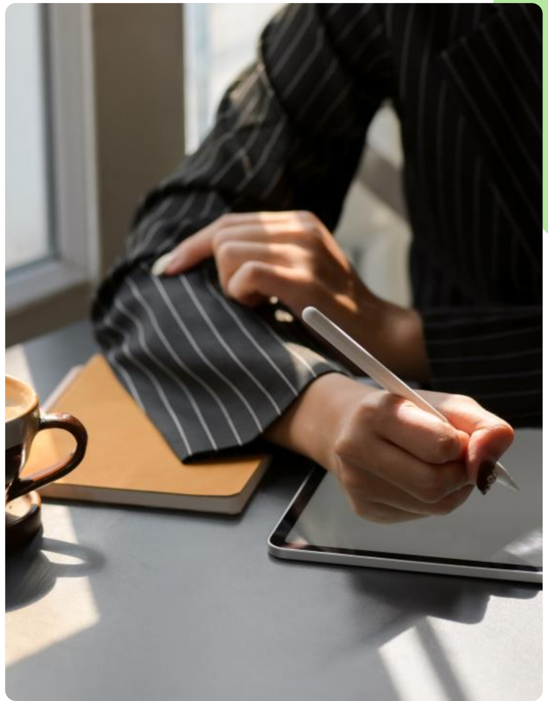
Spread/dispersion

Range, variance, standard deviation, skewness



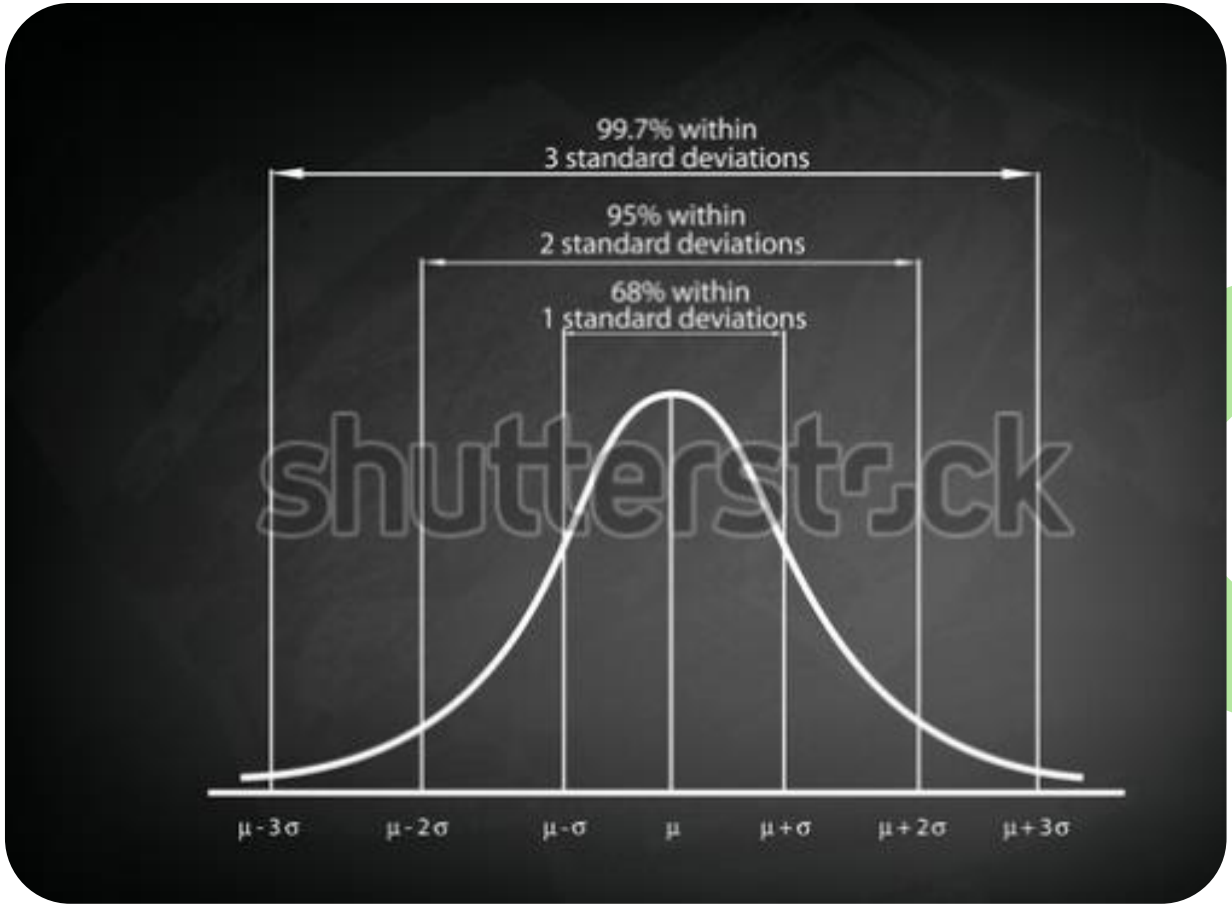
Measures of central tendency

- Mean
- Median
- Mode


Measures of dispersion

- Range
- Variance
- Standard deviation
- Skewness



Measures of dispersion

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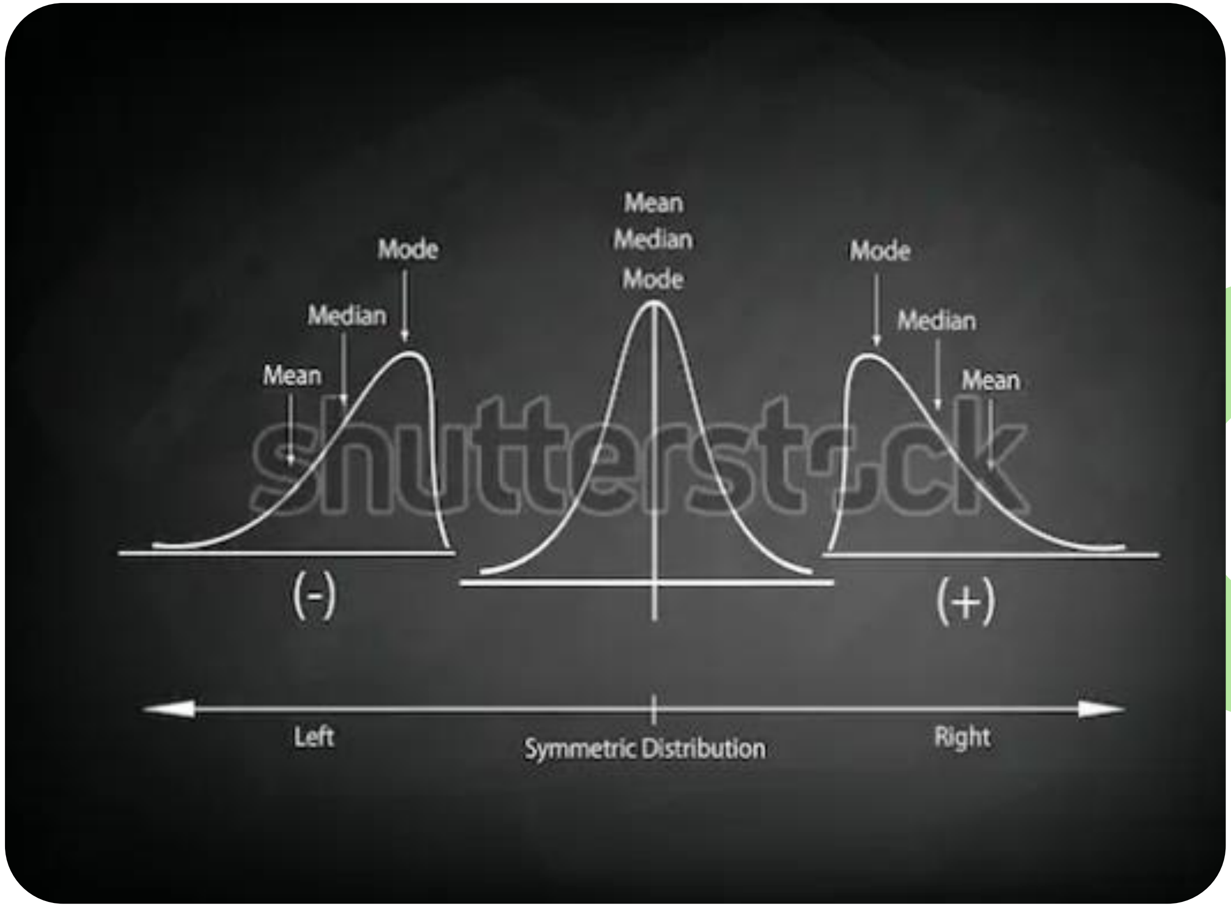


Visualizations

Why visualise?

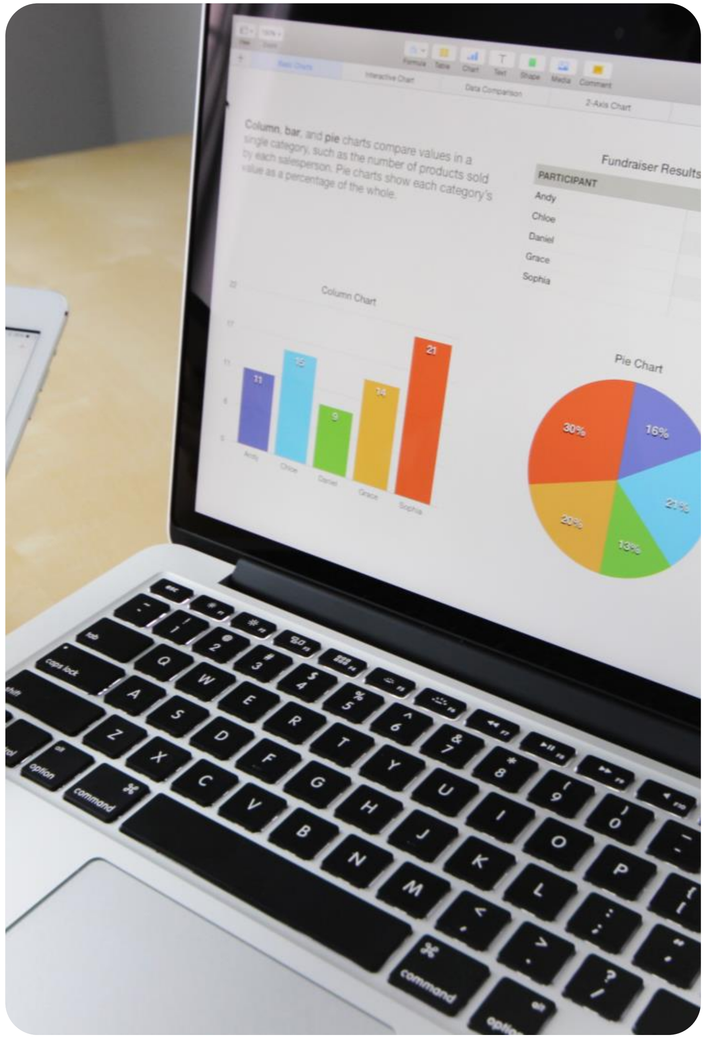
(another way of describing data)

- Sketch a clear picture
- Information easier to understand
- Comparison
- Understand distribution
- Spotting outliers

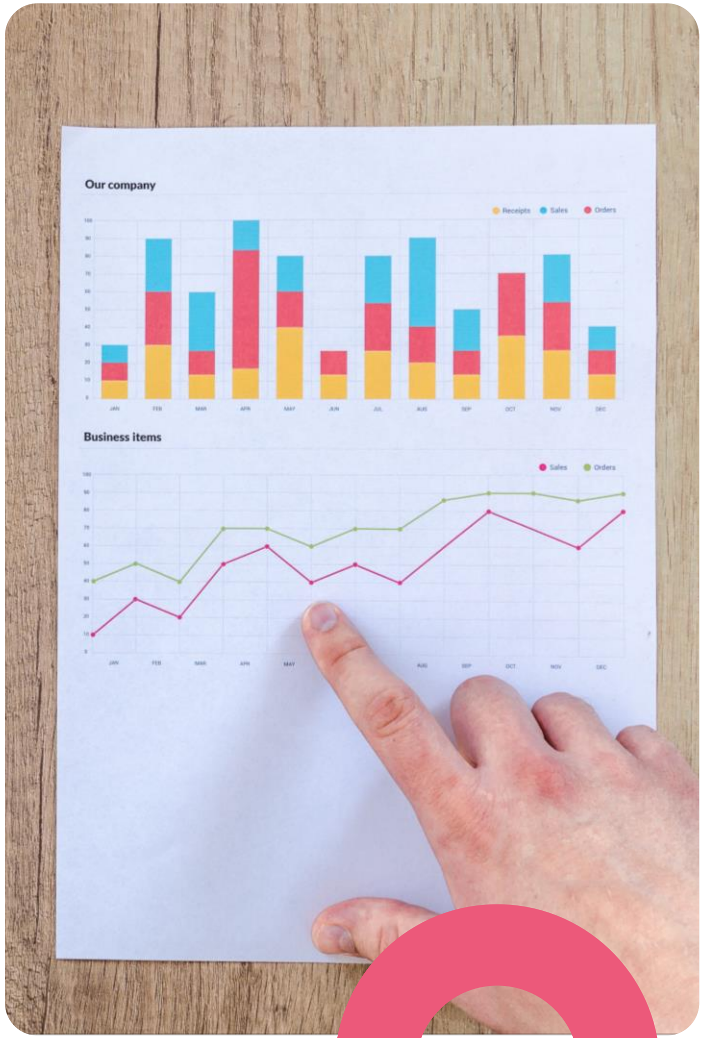
Categorical data visualization

(a look into 2 of the ways of describing data)

Pie chart

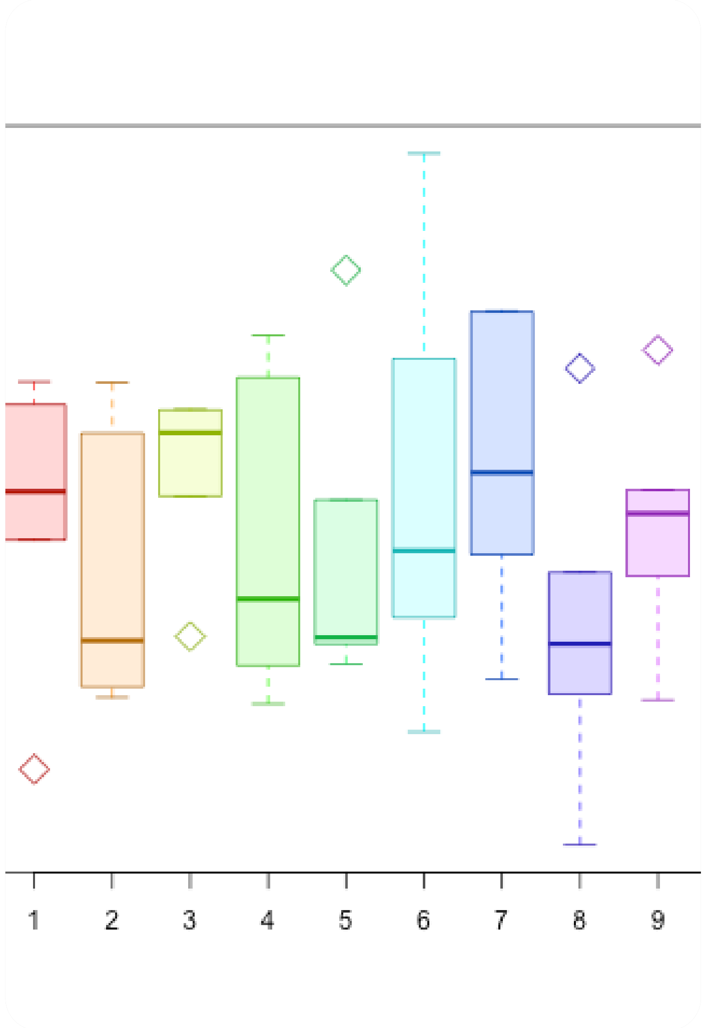


Bar chart

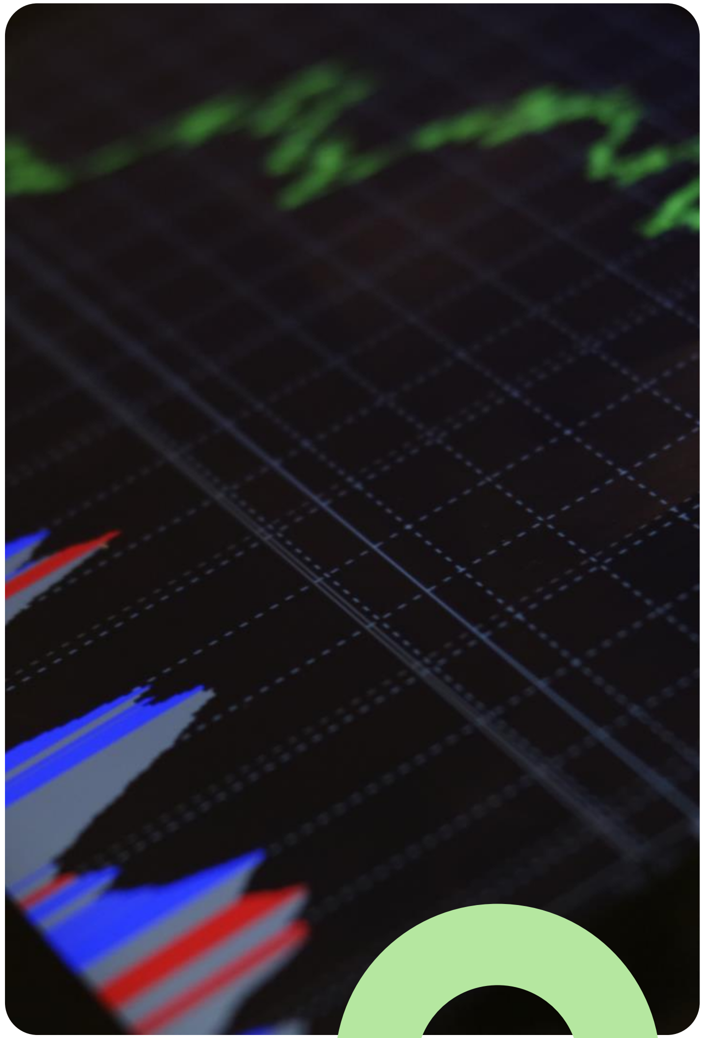


Continuous data visualization

Boxplot



Histogram



Deceiving visualisations

- No baseline
- Manipulating Y-axis or X-axis
- Cherry picking data
- Incorrect visualisation

Challenge

Visualise and describe the remaining variables.

#exploredata